



YASHWANTRAO CHAVAN MAHARASHTRA OPEN UNIVERSITY, NASHIK

(Estd. By Government of Maharashtra)

Dnyangangotri, Near Gangapur Dam, Govardhan, Nashik-422 222

Home Assignment

Academic Year - 2025 - 2026

Name of the Programme & Prog. Code	P131 - B.C.A.(with Credits) - Distance Education - 2016 CGPA - T.Y.B.C.A Semester VI & P131		
Year / Semester	T.Y.B.C.A / Semester VI		
Name of the Course & Course Code	Android Programming (CMP516)		
Name of the Student	KHAN ARSHIYA BANO FERAZ		
Permanent Registration Number	2023017000841576		
Regional Centre	Y.C.M.OPEN UNIVERSITY REGIONAL CENTER MUMBAI & 3139A		
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Name of the Study Centre & Code	K.M.E SOCIETY'S G.M. MOMIN WOMEN'S COLLEGE, BHIWANDI & 35226		
Date of Submission		No of Pages	09

General Instructions

1	Home assignment is useful for self-analysis of your preparation of the final examination and progress
2	Read all the questions and their marks in the home assignment carefully understand the Definitions and concepts properly
3	Read carefully the text, syllabus and summary related to home assignment.
4	Do not copy the points, matter from the text while writing home assignment mentions your one point /opinions whenever necessary
5	Your answer should include how you would apply the knowledge in real life situations
6	Use A4 size lined papers for writing home assignment by taking printout from this file only
7	Write each new answer on a different page.
8	Printed or typed answers are not allowed
9	The marks of Home assignment and corrective instructions will be sent after checking home assignment
10	Present the Home Assignment by following the instructions and rules. Read carefully all the instructions and rules before any correspondence or communication with the university
	For the programs under NEP 2020 Pattern each theory course of 4 credits its carry 30 marks, there are Two Home Assignments:
11	i) Home Assignment 01 (HA01) is of Twenty marks consisting of 2 Short Answer Questions carrying 5 marks each and 1 Long Answer Question of 10 marks.
	ii) Home Assignment 02 (HA02) is of Ten marks consisting of total 10 Objective Type Questions (MCQ) carrying 1 mark each.
12	For each theory course of 2 credits, there is Only One Home Assignment of Fifteen Marks consisting of 1 Short Answer Question of 5 marks and 1 Long Answer Question of 10 marks.
13	Student have to write the Home assignment 02 (MCQ) on separate page. Home Assignment 02 also to be uploaded along with Home assignment 01.



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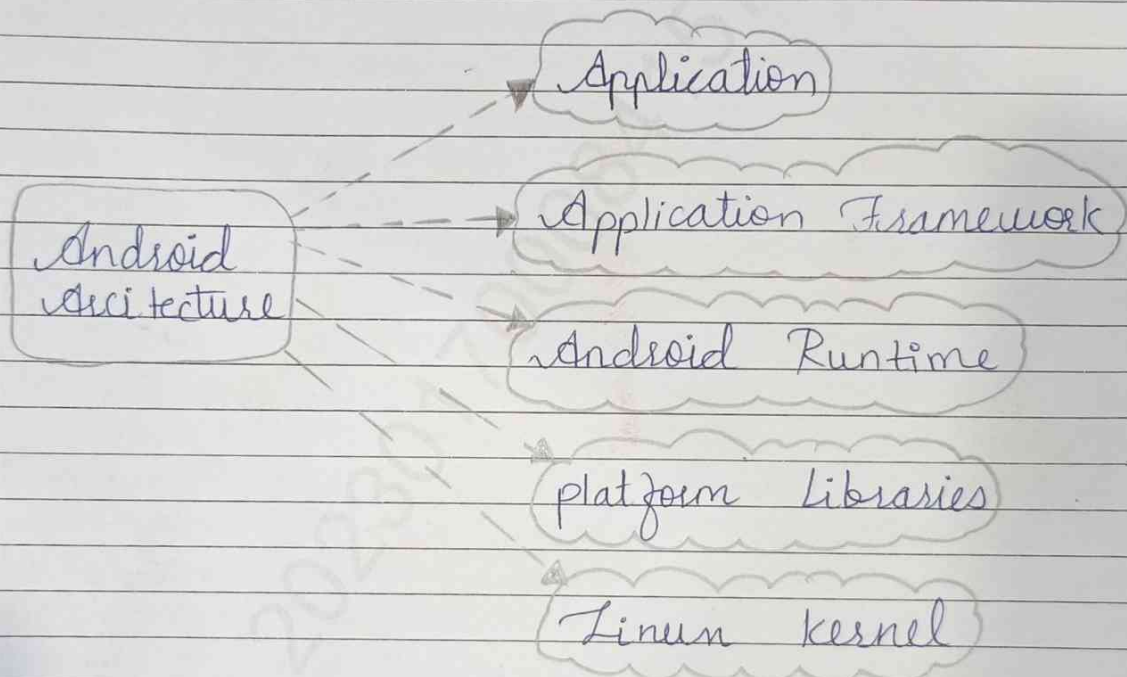
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Q.NO.

Draw Android architecture and explain with all components.

- Definition of Android Architecture is a layered structure that defines how the Android operating system is designed and how different components interact with each other to run applications efficiently.
- It ensures proper communication between hardware and software.



All Five Layers

1) Linux Kernel

- This is the base layer of Android.



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It interacts directly with hardware and manages memory, processes, and device drivers.

2) Native Libraries

- These are C/C++ libraries used for graphics, database (SQLite), and media functions.

3) Android Runtime [ART]

- It executes Android applications.
- It includes core libraries and converts app code into machine code.

4) Application Framework

- provides services like Activity Manager, Window Manager, and Content providers to help developers build apps.

5) Application Layer

- This is the top layer where all user applications are present like contacts, messages, and third-party apps.

Example

- When you open an app (like WhatsApp)

- ① Application layer shows the app
- ② Framework manages UI and activity.
- ③ Runtime executes code
- ④ Libraries handle functions
- ⑤ kernel interacts with hardware



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What is CRUD operations in sqlite database?
Write the query for each operation.

Definition of CRUD

- CRUD stands for Create, Read, Update and Delete.
- It represents the basic operations used to manage data in a database.
- In SQLite, CRUD operations are used to insert, retrieve, modify and remove data from tables.

Table Example

```
CREATE TABLE Students (  
    id INTEGER PRIMARY KEY,  
    name TEXT,  
    age INTEGER  
);
```

Query of each operation

1) Create {Insert Data}

- It is used to add new records into a table.

Eg; `INSERT INTO Students (id, name, age)
VALUES (1, 'Noor', 20);`

2) Read {Select Data}

- It is used to retrieve data from a table.

Eg; `SELECT * FROM Students;`



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3] Update & Modify Data

- It is used to modify existing records in a table.

Eg; UPDATE Students
SET age = 21
WHERE id = 1;

4] Delete (Remove Data)

- It is used to delete records from a table.

Eg; DELETE FROM Students
WHERE id = 1;

- Syntax

INSERT

- INSERT INTO table_name (column1, column2)
VALUES (value1, value2);

SELECT

- SELECT column1, column2 FROM table_name;

UPDATE :

- UPDATE table_name
SET column = value
WHERE condition;

DELETE

- DELETE FROM table_name
WHERE condition;



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Q.NO.		
#	Explain mobility management in GSM with call origination and call termination process.	
#	<u>Definition of GSM</u> • GSM (Global System for mobile communication) is a digital cellular network technology used for transmitting voice and data services through mobile devices.	
#	<u>Definition of Mobility Management</u> • Mobility Management in GSM is the process of tracking and managing the location of mobile users so that they can make and receive calls while moving from one location to another.	
#	<u>Main Components</u>	
①	MS (Mobile Station) • User's mobile device with SIM card.	
②	BTS (Base Transceiver Station) • Communicates directly with the mobile device.	
③	BSC (Base Station Controller) • Controls multiple BTS and manages radio resources.	
④	MSC (Mobile Switching Center): • Handles call routing and switching.	
⑤	HLR (Home Location Register): • Store permanent subscriber information.	



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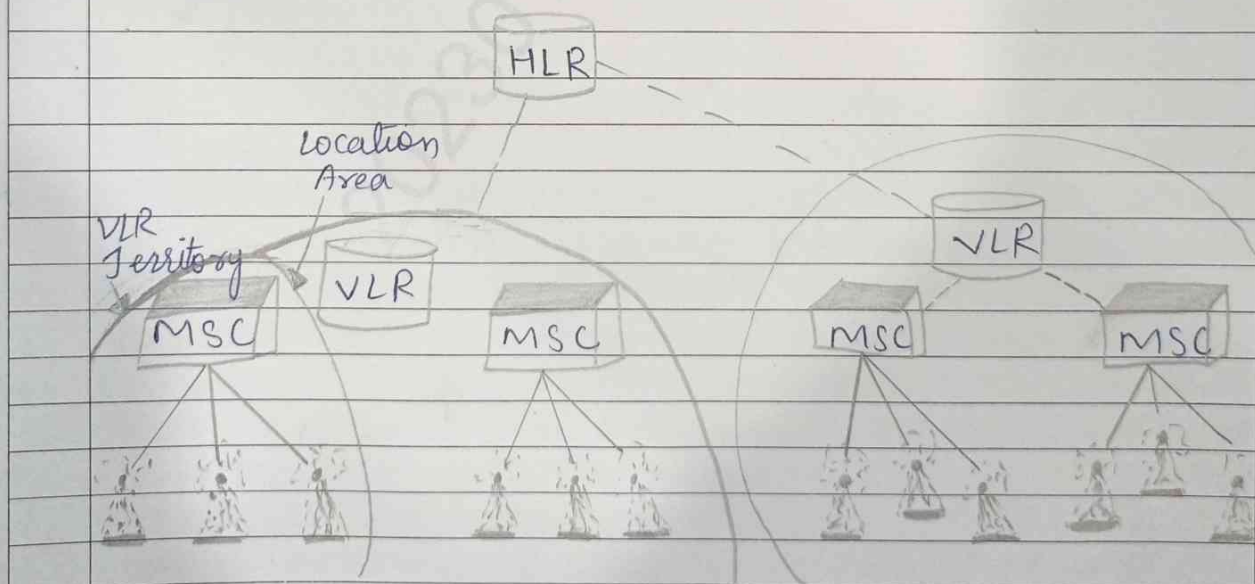
6) VLR (Visitor Location Register):
Stores temporary data of users in a region.

Call Origination process

- ① User dials a number from mobile (MS).
- ② Request is sent to BTS.
- ③ BTS forwards it to BSC.
- ④ BSC sends it to MSC.
- ⑤ MSC checks user details from HLR/VLR.
- ⑥ MSC connects the call to the receiver.

Call Termination process

- ① Incoming call reaches MSC
- ② MSC checks location of receiver using HLR/VLR.
- ③ MSC sends request to BSC.
- ④ BSC forwards it to BTS.
- ⑤ BTS sends signal to mobile (MS).
- ⑥ Receiver answers → call connected.



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#	Define and explain android operating system with at least five applications	
#	<u>Definition of Android OS</u>	
	- Android operating System $\{OS\}$ is an open-source, Linux-based mobile operating system developed by Google. - It is designed for touchscreen devices such as smartphones, tablets and other smart devices, and allows users to run applications and perform various tasks efficiently.	
#	<u>Features of Android</u>	
①	<u>Open Source Nature</u>	
	Android is freely available and developers can modify it according to their needs.	
②	<u>User-Friendly Interface</u>	
	It provides an easy and interactive touch based interface.	
③	<u>Multitasking</u>	
	Users can run multiple applications at the same time.	
④	<u>Connectivity</u>	
	Supports Wi-Fi, Bluetooth, GPS and mobile data.	



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(5) Large App Support
Millions of apps are available on the Google play Store.

(6) Customization
Users can customize themes, widgets, and layouts.

Architecture

• Android architecture is divided into different layers:

(1) Linux kernel :-

Acts as the base layer and manages hardware memory and processes.

(2) Native Libraries :-

provide support for graphic, media and database (SQLite).

(3) Android Runtime (ART) :-

Responsible for executing applications.

(4) Application Framework :-

provides services like activity manager and window manager.

(5) Application Layer :-

Includes all user and system applications.



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Applications of Android1] Smartphones :

- Used for communicating, internet browsing, social media and daily tasks.

2] Tablets :

- It is used for education, reading e-books, watching videos, and productivity work.

3] Smart TVs :

- It is used for streaming online content like YouTube, Netflix etc.

4] Wearable Devices :

- It is used for in smartwatches and fitness trackers for health monitoring.

5] Automobiles & Android Auto :

- It is used for navigation, music and hands-free calling in vehicles.

6] Smart Home Devices :

- It is used for smart speakers, security systems and home automation.

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